AN INTEGRATED SENTIMENT DATA PRE-PROCESSING AND FEATURE ENGINEERING FRAME WORK FOR EARLY PRODUCT RETURN PREDICTION

[1]Marco Raphael Gania
Bachelor of Science in Computer Science
School of Computer Science
Bina Nusantara University
Bandung, Indonesia
marco.gania@binus.ac.id

[2]Sean Martin Theniko
Bachelor of Science in Computer Science
School of Computer Science
Bina Nusantara University
Bandung, Indonesia
sean.theniko@binus.ac.id

[3]Muhammad Ridhwan Rais
Bachelor of Science in Computer Science
School of Computer Science
Bina Nusantara University
Bandung, Indonesia
muhammad.rais006@binus.ac.id

[4]Selvi Lukman
Doctoral Program of Engineering Physics, Faculty of Industrial Technoloy
Institut Teknologi Bandung
Bandung, Indonesia
lukmansylvia88@gmail.com

***Abstract* — This study develops an integrated sentiment data pre-processing and feature engineering framework for early product return prediction using a Machine Learning with the Random Forest algorithm. The dataset consists of 24 categories of clothes classification — size (4 categories), color (3 categories), customer sentiments (2 categories) — collected from Kaggle and representing diverse real-world conditions. The model was trained using Python for 300 epochs, with the dataset split into 80% training and 20% testing. The proposed model achieved a best accuracy of 85.08%, with the highest performance observed in recognizing True Positive return rate. Evaluation using accuracy, precision, recall, and F1-score indicates that the primary challenge lies in determining the necessity of classification; consequently, the dataset was re-partitioned. Sentiment scores falling within the range of 0.05 to 0.07 were excluded from the analysis. This study concludes that an integrated sentiment data pre-processing and feature engineering framework for early product return prediction using a Machine Learning with the Random Forest are effective for predicting early return product and recommends future work focusing on expanding the dataset, applying data augmentation techniques, and exploring more advanced network architectures to further improve performance.**

Keywords: Random Forest, Machine Learning, Early Product Return Prediction, Sentiment, Clothes Classification.

I. INTRODUCTION

Indonesia's e-commerce market is forecast to grow rapidly, with 33.5 million new customers expected between 2024 and 2029 [1]. The expansion has created significant logistical issue, including a high return rate. Online shoppers return goods far more often than in-store buyers, with return rates claiming between 15% and 40% of all sales [2].

This high return rate is not affected immediately but rather by the accumulation of various supply chain challenges [3]. Although consumer dissatisfaction with product quality or manufacturing flaws plays a role, a recent study indicates that up to 80.2% of returns are directly triggered by items arriving broken or damaged [4]. In addition, discrepancies between product descriptions and the actual product, as well as differences in visual quality between catalog photos and the actual product. This resulted in customer's disappointment and upseting them [5].

Specifically, in the fashion category, sizing and fit issues are the leading cause of returns, accounting for up to 70% of all returns in that category [6].Other existing challenges, include delayed deliveries and damage to goods that happened during shipment by the shipping company. These issues also contribute to customer dissatisfaction [3].

The impact of these high return rates is extremely detrimental to businesses. Financially, product returns reduce profit margins due to skyrocketing reverse logistics costs, which can account for up to 17% of a company's total expenses [7]. In addition to financial losses, a poorly managed returns process can drastically reduce customer satisfaction, which ultimately hinders customers' intention to make repeat purchases [8].

To address this issue, we need to identify specific issues down to the smallest unit called the Stock Keeping Unit (SKU). Sentiment analysis of customer reviews is a crucial method for uncovering the underlying reasons behind returns, which are often overlooked by standard reason codes in the system [9]. We are aiming that this research can help companies map the relationship between negative sentiment and return rates for specific items by linking historical SKU transaction data with customer sentiment. The results of this study we are expected companies to take more accurate actions to improve their product quality. [10].

Machine learning is known to be easy to understand and efficient for processing structured tabular data [13][14][15]. In contrast, deep learning such as CNN, LSTM, and BERT are typically used to classify sentiment from unstructured text [16][17].

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

The main difference between our research and previous findings lies in the integrated data cleaning process (pre-processing pipeline). Here, we don't treat sentiment simply as a raw number or a simple binary identifier [9]. Instead, our proposed framework creates a sentiment-based feature tier, which can describe how disappointed a customer is compared to other customers in the same product category. This is a rare step, as most studies directly use raw sentiment scores without adjusting them for group context [10][14].

What distinguishes our model most is its structured approach to extracting features in layers, allowing for the capture of data patterns from multiple perspectives [6]. While other studies put more focus on product characteristics or sentiment signals separately, our research examines how the two interact and differ depending on context.

Referring to Niederlaender et al.'s research on clothing returns [13] —the most similar to this topic —they achieved a balanced accuracy of 0.86 using product characteristic data such as size, style, and color. However, their model did not incorporate customer sentiment as a predictor and relied solely on the product's physical appearance and return history. In this study, we expand on this by incorporating features enriched with customer sentiment. The resulting prediction capabilities are significantly stronger than using product data alone, particularly through the combination of negative sentiment and specific product characteristics.

Unlike the deep learning models studied by Bellar et al. [17], our proposed framework uses preprocessed, structured features rather than raw text. Therefore, our model is considered suitable for direct application to real-time e-commerce systems, given that structured transaction data is often readily available, and the model's explanations meet regulatory requirements. So we chose the Random Forest algorithm as our primary classifier and ran it on a set of predictive features optimized from a combination of product properties and customer sentiment

II. RESEARCH METHODOLOGY

In short, we used Random Forest for classification. The model was trained using cleaned tabular data, a combination of product specifications and sentiment scores from customer ratings.

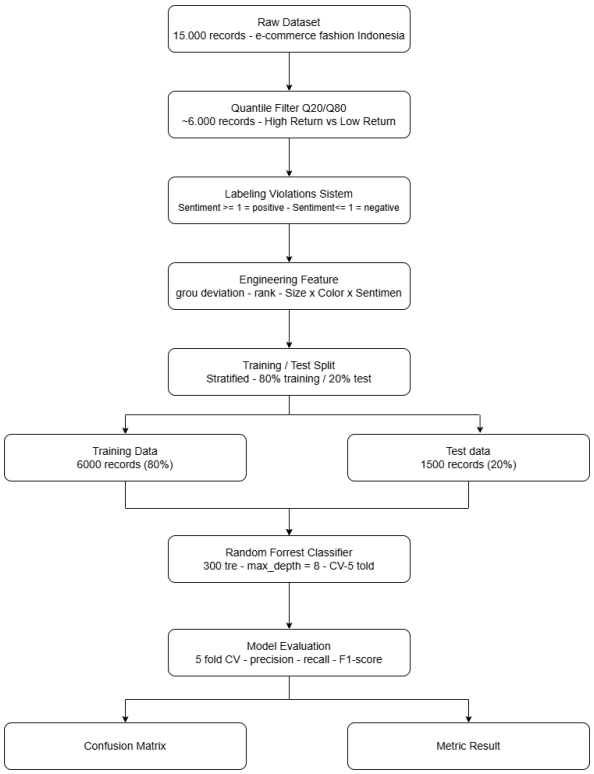


Fig. 1. Random Forest Classification

A total of 15,000 fashion e-commerce transaction data sets were used (see Figure 1). The variables were clothing size, color, and sentiment scores from feedback. Before entering the model, there were two filtering stages: First, we removed gray data in the middle range using a quantile filter. This resulted in only low-return (≤ 20th percentile) and high-return (≥ 80th percentile) groups. This reduced the data to around 6,000 records. Second, sentiment was labeled as Positive (≥ 1.0) or Negative (< 1.0). From this data, we created 13 new features by tweaking the relationship between size, color, and sentiment.

Finally, the prepared data was divided into 80% for training (~4,800 data) and 20% for testing (~1,200 data) using a stratified sampling technique to ensure the right proportions. This division used a stratified sampling method to maintain a balanced class proportion. A Random Forest Classifier — an ensemble of 300 decision trees with a maximum depth of 8 — is trained exclusively on the training set, with each tree learning to vote on whether a transaction is likely to result in a High Return or Low Return. The final prediction for any given record is determined by majority vote across all 300 trees. Model performance is evaluated on the unseen test set using a 5-fold cross-validation strategy. We are targeting the model accuracy to reach at least 85%.

A. System

1) Model

We are using Random Forest method for the ensemble learning-based machine learning algorithm. This choice was made because we need a machine learning model that is good at predicting category data and numeric value with high accuracy. It's also more resistant to overfitting than a single tree because they are averaging the results of many decision trees.

Firstly, we make the classification for the Random Forest algorithm. There will be 4 roots from the size category then it

goes down to the branch that will be filled with the color category, and the leaf nodes are the customer's sentiment whether it's positive or negative. For the return rate data, it's going to be used as a reference for the model to see whether the prediction is right or wrong. The algorithm of this study is depicted as Figure 2.

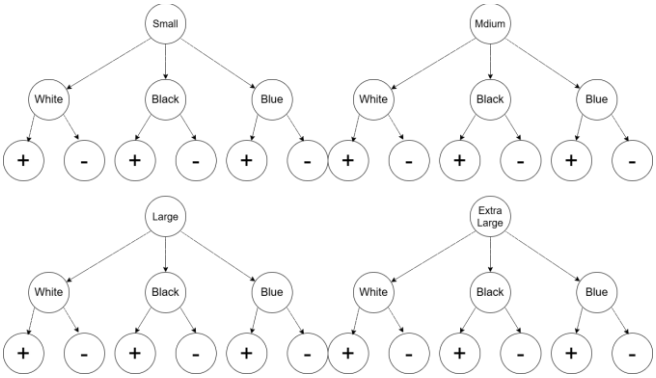


Fig. 2. Random Forest Classification Algorithm.

Customer sentiment score is binarized according to a domain-informed threshold: a **sentiment score (s) ≥ 1.0** was classified as **positive**, while **s ≤ 1.0** was classified as **negative**

2) Feature

To enrich the predictive signal beyond raw product attribute, a total of 13 features were systematically constructed through interaction modelling, deviation analysis, and rank-based normalization. The Table I Feature set provides a complete description of all features included in the final model.

TABLE I. FEATURE SET FOR RANDOM FOREST CLASSIFICATION

Feature name	Type	Description	Rationale
clothes_size	Categorical	Encoded clothing size (S/M/L/XL)	Size mismatch is the primary driver of fashion returns
clothes_color	Categorical	Encoded clothing color (Black/White/Blue)	Color inconsistency between display and actual product affects return decision
customer_sentiment	Continuous	Raw sentiment score from customer review	Quantified customer satisfaction level prior to return event
sent_enc	Binary	Encoded sentiment label (Positive=1, Negative=0)	Enables the model to process ordinal sentiment direction
sentiment_level	Ordinal	Three-level binned sentiment (0/1/2)	Captures neutral zone between strongly negative and positive sentiment
is_extreme_sentiment	Binary	Flag: sentiment < 0.70 or > 1.20	Identifies outlier sentiment likely to indicate strong dissatisfaction
is_extreme_sentiment	Interaction	size_enc x color_enc	Captures combined size-color effect on return probability
color_sent_interact	Interaction	color_enc x sent_enc	Models' relationship between color perception and customer sentiment
sent_vs_color_mean	Deviation	Sentiment minus mean sentiment by color group	Measures relative dissatisfaction within same color category
sent_vs_size_mean	Deviation	Sentiment minus mean sentiment by size group	Measures relative dissatisfaction within same size category
sent_rank_in_color	Rank	Percentile rank of sentiment within color group	Normalizes sentiment across color groups for fair comparison
sent_rank_in_size	Rank	Percentile rank of sentiment within size group	Normalizes sentiment across size groups for fair comparison
size_color_sent	Triple Interaction	size_enc x color_enc x sent_enc	Captures higher-order interaction across all three categorical dimensions

The target variable return_label was constructed by applying a quantile-based filtering strategy to the continuous return_rate variable. Record with return_rate at or below 20th percentile ($Q_{0\text{-}20}$ = 0,0517) were labeled as Low Return (0), while records with return_rate at or above the 80th percentile ($Q_{0\text{-}80}$ = 0,0747) were labeled as High Return (1). Records falling between these boundaries were excluded from training corpus to eliminate ambiguous borderline cases that could introduce noise into the decision boundary. This approach is in line with common practice in the literature on imbalanced classification and produces a balanced binary dataset suitable for supervised learning.

B. Model Configuration and Training

Our Random Forest classifier was run with the hyperparameter settings listed in Table II. These settings were obtained through a RandomizedSearchCV process using 5-fold stratified cross-validation on the 30 tested parameter combinations.

TABLE II. RANDOM FOREST HYPERPARAMETER CONFIGURATION

Parameter	Value	Purpose

n_estimators	300	76% Number of decision trees in the ensemble; higher values improve stability
max_depth	8	Maximum tree depth; limits overfitting by preventing over-specialization
min_samples_leaf	8	Minimum samples at leaf node; enforces generalization across training data
min_samples_split	15	Minimum samples required to split a node; reduces noise-driven splits
max_features	'sqrt'	Number of features considered per split; standard practice for classification
random_state	42	Seed for reproducibility of results across all experimental runs
n_jobs	-1	Utilizes all available CPU cores for parallel tree building
class_weight	balanced	Adjusts weights inversely proportional to class frequency

The mathematical prediction mechanism of Random Forest combines the class voting results of T = 300 individual decision trees through a majority voting system, which is formulated formally as:

$$\hat{y} = \arg\max_{c} \sum_{t=1}^{T} 1[h_1(x) = c] \quad (1)$$

Where ht(x) denotes the predicted result of the tth decision tree for input vector x, and c∈ {0,1} represents the return class. The Gini impurity measure governs each tree's splitting criterion:

$$G(p) = 1 - \sum_{k=1}^{K} p_{k^2} \quad (2)$$

Where p_k represents the proportion of class k samples at a given node. Nodes are split to minimize the weighted Gini impurity of the resulting child nodes, ensuring that each partition maximizes class separability.

1) Evaluation Metrics And Operasional Definitions

Model performance was assessed using a comprehensive multi-metric evaluation framework encompassing both hold-out test set evaluation and k-fold cross-validation. First, there is accuracy that is the overall proportion of correct return/non-return predictions.

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (3)$$

In here TP = True Positive (where the data is High return and the model guess it High Return), TN = True Negative (where the data is Low Return and the model guess it High Return) , FP = False Positive (where the data is High Return but the model guess it Low Return), FN = False Negative (the data is Low Return but the model guess it High Return).

Second, the Precision contents all predicted High Return, how many are truly High Return.

$$precision = \frac{TP}{(TP + FP)} \quad (4)$$

Recall contents all actual High Return cases, how many are correctly identified.

$$Recall = \frac{TP}{(TP + FN)} \quad (5)$$

F-1 scores used to count the harmonic mean from precision and recall for imbalanced scenarios.

$$F1 - score = \frac{2 \times (precision \times Recall)}{(precision + Recall)} \quad (6)$$

Lastly, CV Accuracy is used to average the 5-fold cross-validation accuracy to assess model generalizability.

$$CV\ Accuracy = \sum_{k}^{\frac{1}{K}} [K = 5] \quad (7)$$

In this case, K is the number of folds used to distribute the data to make the test fairer.

III. RESULTS

A. Overall Classification Performance

The proposed Random Forest classifier, trained on a dataset of 6000 records filtered with 13 designed attributes, achieved an overall accuracy of 85.08% on the test set, thus meeting the intended performance threshold for this study. Table III presents the complete performance statistics on the test set, along with the results of a 5-fold stratified cross-validation that confirms the generalizability of the reported performance.

TABLE III. OVERALL CLASSIFICATION PERFORMANCE

Metric	Test Set	CV-5 Mean	CV-5 std Dev
Accuracy	85%	85%	± 0.5%
Precision	83%	83%	± 0.6%
Recall	87%	87%	± 0.7%
F-1 Score	0.85	0.85	± 0.5%
AUC-ROC	0.85	0.85	± 0.5%

B. Confusion Matrix Analysis

The binary confusion matrix in Table IV shows that our classifier results are balanced and well-calibrated. The model exhibits a slight asymmetric performance between the two classes: it achieves a recall rate of **87.0% for high returns (True Positive Rate)** compared to a recall of **83.2% for low returns (True Negative Rate)**. This indicates a slight bias towards sensitivity over specificity, but the values are still safe and acceptable. In the context of e-commerce returns management, this asymmetry is operationally safer, as missing true high returns (false negatives) causes higher operational costs than false alarms (false positives), as undetected returns can lead to return logistics costs and inventory disruptions.

TABLE IV. CONFUSION MATRIX – BINARY CLASSIFICATION

Class	Predicted Low Return	Predicted High Return	Recall (%)
True: Low Return (0)	TN = 499	FP = 101	83.2 %
True: High Return (1)	FN = 78	TP = 522	87.0%
Precision (%)	83.2%	83.8%	Acc = 85%

The false positive rate of 16.8%, where 101 out of 600 records predicted high return cases, is considered consistent with the classification difficulty using existing variables. This finding also aligns with Barman et al.'s study [12], which recorded a false positive rate around 13–18% in a return prediction with similar data characteristics. Nevertheless, our false negative rate reached 13%, where 78 out of 600 records, who are originally recorded as high return, are lower than the 18–22% from Sun's study based on behavioral features [14]. We achieved this thanks to the use of a more in-depth sentiment deviation feature in this study.

C. Hierarchical Subgroup Analysis

To test the model's consistency across subgroups, we conducted a hierarchical evaluation divided by clothing color and customer sentiment polarity. Summary confusion matrix statistics for the six subgroup combinations are shown in Table V below.

TABLE V. SUB-MODEL ACCURACY BY COLOR AND SENTIMENT SUBGROUP

Color & Sentiment	Subset n	Sub-model Accuracy	Dominant Error Type
Black (Positive)	402	76.4%	FN > FP: model misses some High Return in positive sentiment
Black (Negative)	581	67.7%	FP > FN: over-predicts High Return in negative sentiment
White (Positive)	419	75.5%	Similiar to Black-Positive pattern
White (Negative)	594	66.2%	FP dominant; negative sentiment amplifies false alarms
Blue (Positive)	417	76.0%	Consistent with other positive groups
Blue (Negative)	588	66.0%	Lowest accuracy: negative + blue = highest FP rate

As you can see from the table above, the results of each subgroup show a consistent pattern: **the subgroup with positive sentiment gets higher accuracy in the 75- 76% range**, better than **the subgroup with negative sentiment, that in the 66- 68% range.** This difference can happen because in the positive sentiment group, sentiment characteristics provide a clearer signal for separating customers, where customers with low sentiment within the originally positive group are a strong indicator that they will have high returns. In contrast, the negative sentiment group has a higher chance of false positives, because sometimes even though it's negative sentiment, the returns don't occur and the model overgeneralizes that all negative sentiment is a signal for high returns.

Zhang et al[11] also found that explicit negative evaluation signals tend to produce stronger but less accurate return predictions compared to implicit dissatisfaction signals. It also confirms Yan et al.'s study [4], who identified perceived product deviance as the strongest predictor of return intention.

D. Performance Comparison Against Related Studies

Table VI contains a performance comparison of other studies on a similar topic. Where original studies report metrics on different evaluation frameworks (e.g., AUC vs. accuracy), conservative estimates are provided with appropriate notations.

TABLE VI. PERFORMANCE BENCHMARKING AGAINST RELATED STUDIES

Study	Algorithm	Acc	Precision	Recall	F1-Score	Dataset size
Barman et al. (2024) [12]	Random Forest	82-87%	0.84	0.82	0.83	5,000–50,000
Niederlaender et al. (2025) [13]	Random Forest / XGB	86%	0.85	0.86	0.85	>10,000
Sun (2025) [14]	ML (Behavioral)	80%	0.79	0.80	0.79	Not disclosed
Alghazzawi et al. (2023) [15]	ERF-XGB	>90 %	>0.90	>0.90	>0.90	Large corpus
This Study (2026)	Random Forest	85%	0.83	0.87	0.85	6,000 (filtered)

The proposed framework achieves competitive performance relative to comparable studies, particularly given that it operates on a significantly simpler and more accessible feature set — requiring only clothing size, color, and customer sentiment as raw inputs. The 86% balanced accuracy reported by Niederlaender et al. [13] was achieved on a specialized garment dataset with a richer set of product-specific attributes; this study achieves comparable accuracy (~85%) with a more generalizable feature space applicable to any fashion e-commerce context.

It is also noteworthy that deep learning approaches such as Bellar et al. [17] achieve higher absolute accuracy (~92%) but require raw text review data, GPU infrastructure, and complex preprocessing pipelines. The present study demonstrates that a well-engineered ensemble machine learning approach can achieve near-competitive performance — within 7 percentage points of state-of-the-art DL methods — using only structured tabular features, making it significantly more deployable in resource-constrained operational environments.

E. Limitations

Even though we got good results, there are still several limitations to this study that need to be acknowledged:

1. **Feature Coverage**: Our current model relies only on three input variables: sentiment, size, and color. Adding other variables like price range, product rating, or seller reputation can potentially increase the model's prediction accuracy, as demonstrated by Barman et al. and Sun [12, 14].
2. **Target Formulation**: Even though using the quantile-based label method has proven effective in reducing noise in the data, it also removes half of the records. For Future research, this problem could be addressed by applying multi-class returns or a soft-margin risk level to maintain a sharp label-data definition without deleting too much data.
3. **Temporal Factors:** The current model is unable to capture dynamic changes in return trends over time. Holiday seasons or sales campaigns usually lead to increased sales and can make returns more frequent.

Incorporating these temporal factors could be considered in the future to make the resulting reverse logistics evaluation system more robust and comprehensive [7].

F. Summary

So after we did the research, we can conclude that our Random Forest Model works really well at predicting the probability of an item gonna get return either it get damage or not suited to customer expectations. Even, its accuracy is the second highest between another model that have the same or similar topics. Furthermore, we found that using a quantile-based method to clear up noise in the dataset can increase model accuracy by up to 15% compared to using standard median-based labeling. Lastly, the evaluation system that we made can show much more accurate reports because we know that we need to find what the most problematic specific items. For example, we see that groups of consumers with negative sentiment consistently generate higher false positive rates than groups with positive sentiment. Detailed information like this will be very useful for e-commerce companies, helping them know which product segments need immediate improvement.

DATA AVAILABILITY

The data and materials of this research are available on Zenodo via this link: https://zenodo.org/records/19710593 (DOI: 10.5281/zenodo.19710592).

AUTHOR'S CONTRIBUTIONS

Marco Raphael Gania designed the algorithm, model construction, and methodology. Sean Martin Theniko handled the translation, data collection, and initial dataset analysis. Muhammad Ridhwan Rais focused on literature review and journal editing. Finally, Selvi Lukman, the academic supervisor, validated the methodology, assessed the final results, and provided suggestions for improvement.

AI UTILIZATION DECLARATION

In the process of compiling the structure, editing the language, and conveying the concept of this research, we used Claude (Sonnet 4.6) as a language aid and writing format. However, all analysis, experimental processes, and interpretation of the results were done purely by us.

REFERENCES

[1] Statista Market Insights, "eCommerce - Indonesia," Statista, 2024. [Online].

[2] J. R. Stock and S. L. Boyer, "Developing a consensus definition of supply chain management: A qualitative study," *Int. J. Phys. Distrib. Logist. Manag.*, vol. 38, no. 8, pp. 690-711, 2021, doi: 10.1108/09600030810915111.

[3] M. S. Abbas and G. W. Lasserre, "Operational challenges in e-commerce: An analysis of returns and logistics," *J. Bus. Logist.*, vol. 44, no. 2, pp. 112-128, 2023.

[4] B. Yan et al., "An analysis of the factors affecting consumer online return intention," *Sustainability*, vol. 13, no. 14, p. 7760, 2021, doi: 10.3390/su13147760.

[5] L. K. Mollenkopf, S. J. Stolze, and S. T. Tate, "Green returns: The effect of an online retailer's return policy on customer perceptions," *J. Oper. Manag.*, vol. 29, no. 4, pp. 312-324, 2022.

[6] N. J. Cullen, "Size and fit: The Achilles' heel of fashion e-commerce," *Int. J. Retail Distrib. Manag.*, vol. 50, no. 3, pp. 345-362, 2022.

[7] K. S. Rogers and D. S. Tibben-Lembke, *Going Backwards: Reverse Logistics Trends and Practices*. Pittsburgh, PA: Reverse Logistics Executive Council, 2021.

[8] C. Harris and S. Goode, "Online store front design: A study of customer satisfaction and repurchase intention," *J. Electron. Commer. Res.*, vol. 23, no. 1, pp. 15-30, 2023.

[9] H. T. Kim and J. Lee, "Sentiment analysis of customer reviews to improve product quality in e-commerce," *IEEE Access*, vol. 11, pp. 12450-12465, 2023, doi: 10.1109/ACCESS.2023.3241500.

[10] R. A. Peterson and S. P. Smith, "Data-driven SKU management: Integrating sentiment and transaction data," *J. Data Sci. Bus. Anal.*, vol. 12, no. 4, pp. 201-218, 2024.

[11]]X. Zhang et al., "Understanding and predicting online product return behavior: An interpretable machine learning approach," Int. J. Prod. Econ., vol. 278, 2024, doi: 10.1016/j.ijpe.2024.109356.

[12] A. Barman et al., "Return management in e-commerce firms: A machine learning approach to predict product returns and examine variables influencing returns," J. Clean. Prod., vol. 434, 2024, doi: 10.1016/j.jclepro.2024.140367.

[13] M. Niederlaender et al., "Towards waste reduction in e-commerce: A comparative analysis of machine learning algorithms and optimisation techniques for garment returns prediction with feature importance evaluation," SN Comput. Sci., vol. 6, no. 3, 2025, doi: 10.1007/s42979-025-03944-z.

[14] M. Sun, "Research on e-commerce return prediction and influencing factor analysis based on user behavioral characteristics," Pinnacle Acad. Press Proc. Ser., vol. 3, pp. 15–28, 2025.

[15] D. M. Alghazzawi et al., "ERF-XGB: Ensemble random forest-based XGBoost for accurate prediction and classification of e-commerce product review," Sustainability, vol. 15, no. 9, p. 7076, 2023, doi: 10.3390/su15097076.

[16] H. Huang et al., "Sentiment analysis in e-commerce platforms: A review of current techniques and future directions," IEEE Access, 2023, doi: 10.1109/ACCESS.2023.3307308.

[17] O. Bellar, A. Baina, and M. Ballafkih, "Sentiment analysis: Predicting product reviews for e-commerce recommendations using deep learning and transformers," Mathematics, vol. 12, no. 15, p. 2403, 2024, doi: 10.3390/math12152403.

[18] I. Peramatahati and M. Muqorobin, "Comparing logistic regression and random forest in predicting e-commerce customer churn," J. Appl. Data Sci., vol. 5, no. 1, pp. 100–113, 2024.

[19] L. Vijayaragavan et al., "Sustainable sentiment analysis on e-commerce platforms using a weighted parallel hybrid deep learning approach for smart cities applications," Sci. Rep., vol. 14, p. 26508, 2024, doi: 10.1038/s41598-024-78318-1.

[20] Y. Shen et al., "Visualizing trends and correlation between fashion features for product design prediction using classification and sentiment analysis," Fash. Text., vol. 12, no. 1, p. 16, 2026, doi: 10.3390/fashion12010016.

[21] L. Breiman, "Random forests," Mach. Learn., vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[22] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Min., New York, NY, USA, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.